Rahul Makwana

Founding Full-Stack AI Engineer

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SUMMARY

Founding Full-Stack AI Engineer with 10 years of experience across enterprise systems, IoT/cloud platforms, and applied AI products. At Tellen, shipped audit and accounting AI products from prototype to production, including financial-statement parsing, footnote workflows, substantive testing, and audit-file completeness tools used across 9–10 clients in pilot and production. Built retrieval-augmented generation (RAG) workflows, multi-agent orchestration patterns, and context/memory strategies for document-heavy AI applications, while improving scalability, performance, and observability through containerized infrastructure, release ownership, and production-grade validation.

EXPERIENCE

Tellen — Remote
Jan 2025 – Present

Founding Full-Stack AI Engineer

- Built and shipped audit and accounting AI products from prototype to production, including Footnote Repository, Internal Consistency Check, Substantive Testing App, and Audit File Completeness, supporting 9–10 clients across pilot and production engagements.
- Developed core workflows using Python, FastAPI, Next.js, React, PostgreSQL, OpenAI APIs, Azure Blob Storage, and Azure Document Intelligence for OCR, document parsing, evidence review, and structured audit workflows.
- Designed retrieval-augmented generation (RAG) and financial document intelligence workflows that grounded model outputs in source documents, extracted evidence, and improved trust, traceability, and reviewability for audit use cases.
- Implemented context and memory strategies for multi-step AI workflows, including structured prompt state, reusable document context, and human-in-the-loop validation checkpoints to improve consistency across long-running review tasks.
- Orchestrated multi-agent workflows across Cursor, Claude Code, ticketing, code review, and release processes to accelerate prototyping, bug fixing, and feature turnaround while maintaining engineering judgment and production accountability.
- Stepped into broad engineering execution ownership within the first week after the CTO departed, coordinating 4 offshore developers plus rotating U.S. contractors, breaking business requirements into technical tasks, reviewing code, and deciding what was ready to ship.
- Owned weekly releases end-to-end, including merge approvals, release notes, deployment monitoring, production bug fixes, and validation of client environments after release; helped improve scalability, performance, and observability as the platform evolved toward Docker and Kubernetes.

Antylia Scientific — Webster, TX
May 2021 – Dec 2024

Senior Software Developer / Team Lead / AI-ML Lead

- Designed and built a calibration management platform from scratch that became operationally critical across the business, supporting 40+ staff and contributing to 43% cost reduction and 33.33% production improvement.
- Served as a key technical owner for TraceableLIVE, a cloud monitoring platform for connected devices, alarms, dashboards, and compliance-oriented workflows, supporting the full flow from device communication to cloud ingestion, alerting, and user-facing application behavior.
- Led NiceLabel-related manufacturing and labeling initiatives across 3 facilities, improving throughput by 40% while strengthening traceability and compliance support.

- Led an offshore team of 4 developers and 2 testers, mentored junior engineers and an intern, and acted as the primary technical escalation point for complex reverse-engineering and customer issues.
- Built applied AI initiatives including an Azure-based customer service chatbot and predictive analytics for TraceableLIVE, while maintaining zero downtime on critical production software over 5 years.

Antylia Scientific — Webster, TX May 2016 – May 2021

Software Developer

- Built internal business applications and automation tools using C#, .NET, SQL Server, and Azure to improve operational visibility and manufacturing workflows.
- Supported cloud-connected product development, integrations, reporting systems, and customer-facing software across regulated and device-driven environments.
- Contributed to software used across the physical product lifecycle, from production operations to go-to-market support, while gaining deep ownership of cross-system troubleshooting and platform behavior.

SKILLS

Languages: Python, TypeScript, SQL, C#

Software Engineering: FastAPI, Next.js, React, Flask, REST APIs, Prisma

Applied AI / LLM Systems: LLM Applications, Multi-Agent Orchestration, RAG, Retrieval-Augmented Generation, Memory Strategies, Context Window Management, NLP, Generative AI, Prompt Engineering, OpenAI APIs, Azure Document Intelligence, OCR, Document Parsing, AI Workflow Automation, Financial Document Intelligence, Chatbot Development, LangChain, TensorFlow, PyTorch, Scikit-learn, Predictive Analytics

Cloud & Data: PostgreSQL, SQL Server, Azure Blob Storage, Azure, Git, CI/CD, Docker, Kubernetes, Scalability, Performance Optimization, Observability, Logging, Monitoring

Leadership & Delivery: Product Development, Release Management, Code Review, Client Requirements Translation, Offshore Team Coordination

EDUCATION

IBM Nov 2024 – Jan 2025

IBM AI Developer Professional Certificate

UT Austin Jan 2022 – May 2023

Post-Graduate Program in AI and Machine Learning

Texas State University Jan 2014 – Dec 2015

M.S. Computer Science